

First Mid-Semester Exam PHY301 (20 marks)

mean =

median = 9.5

1. Consider the circuit equation for current I in a LCR circuit:

(STABLE)

 $R^2 C < 4L$

spiral

 $R^2 C > 4L$

Nodes

 $R^2 = 4L$

degenerate node

$$L\ddot{I} + R\dot{I} + I/C = 0$$

Given $L, C > 0$, what is the nature of the fixed point(s) as R is varied ($R \geq 0$).

(5 marks)

2. Consider a conservative Newtonian system where a particle of mass $m = 1$ moves in a potential

$$V(x) = \frac{1}{2}x^2 - \frac{1}{4}x^4$$

- Enumerate the equilibrium (fixed) point(s) of the system.
- Find the nature of the fixed point through analysis of the Jacobian.
- Draw the phase portrait of the dynamics, and mark the separatrix.
- What are the distinct types of motion that can take place in the system?

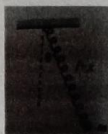
(8 marks)

3. Consider a system given by the equation: $\ddot{x} + 2\lambda\dot{x} + \omega^2x = F$, where F is a constant.

Find the solution $x(t)$ for the system, for the case of $\lambda = \omega$, with initial condition $x(0) = 0$, $\dot{x}(0) = 1$.

(7 marks)

1. Consider a pendulum made of a spring with mass m on the end. The spring is arranged to lie on a straight line (which we arrange by, say, wrapping the spring around a rigid massless rod). The equilibrium length of the spring is l . Let the spring have length $l + x(t)$ and let the angle with the vertical be $\Theta(t)$. Assume the motion takes place in a vertical plane.



- (a) Write down the Lagrangian.
 (b) Write down the Euler-Lagrange equations.
 (c) Find the equations of motion for x and Θ .

(6 marks)

2. Given a point mass m with Lagrangian of the form:

$$L(r, \theta, \dot{r}, \dot{\theta}) = \frac{m}{2}(\dot{r}^2 + r^2\dot{\theta}^2) + \frac{k}{r}$$

Find the Hamiltonian using Legendre Transforms.

(5 marks)

3. Show that

$$\frac{\partial^2 H}{\partial p^2} \cdot \frac{\partial^2 L}{\partial u^2} = 1$$

(3 marks)

4. A particle moves in a potential

$$V(r) = -V_0 e^{-(\lambda^2 r^2)}$$

- (a) Write down the effective potential for the system, for a given angular momentum ℓ .
 (b) Given ℓ , find the radius of the stable circular orbit. An implicit equation is fine.
 (c) It turns out that if ℓ is too large, then no circular orbit exists. What is the largest value of ℓ for which a circular orbit does in fact exist?

(6 marks)